



Junior Physics Olympiad

Inspiring Young Minds to Explore the World of Physics

Class : X Syllabus

UNIT 1: Advanced Kinematics

- Relative motion
- Projectile motion
- Vector resolution
- Two-dimensional motion

UNIT 5: Electricity

- Electric current
- Potential difference
- Resistance
- Ohm's Law
- Resistivity
- Series and parallel circuits
- Electrical power
- Heating effect

UNIT 2: Advanced Dynamics

- Newton's Laws
- Friction
- Circular motion
- Centripetal force

UNIT 6: Magnetism and Electromagnetism

- Magnetic field
- Field lines
- Electromagnets
- Electric motor
- Electromagnetic induction
- Generator

UNIT 3: Rotational Motion

- Torque
- Moment arm
- Centre of mass
- Stability
- Equilibrium

UNIT 7: Optics

- Reflection
- Refraction
- Mirrors
- Lenses
- Human eye
- Optical instruments
- Total internal reflection

UNIT 4: Gravitation

- Orbital motion
- Escape velocity
- Artificial satellites

UNIT 8: Modern Physics

- Structure of atom
- Radioactivity
- Nuclear energy
- Space science
- Astronomy basics